

PROJECT REPORT

Title: The project Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

1. INTRODUCTION

1.1 Project Overview

The project Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer aims to replace traditional manual procurement processes in ServiceNow with an automated workflow. By leveraging ServiceNow Flow Designer, standard laptop requests are automatically processed, creating catalog tasks for hardware configuration and routing them to the appropriate fulfillment groups without human intervention.

1.2 Purpose

The primary purpose of this project is to reduce manual overhead, eliminate delays caused by manual task creation, minimize human error, and provide an efficient, transparent procurement experience for both employees and the IT hardware team.

2. IDEATION PHASE

2.1 Problem Statement

The current IT procurement process lacks efficiency and automation, resulting in delays and manual overhead, particularly in handling standard laptop orders. Requests for standard laptops often require configuration, but this step is prone to oversight or delay, leading to frustration among users and inefficient resource allocation within the IT department.

2.2 Empathy Map Canvas

- **Says:** "Ordering a laptop takes too long and I don't know the status."
- **Thinks:** "IT procurement should be automated and hasslefree."
- **Does:** Searches the ServiceNow portal, submits a request, and follows up manually with IT.
- **Feels:** Frustrated by delays and lack of real-time updates.

2.3 Brainstorming

During the brainstorming session, three main approaches were evaluated:

1. **Manual Notifications:** Sending emails to IT upon request submission (Rejected due to high delay/error risk).
2. **Scripted Workflows:** Writing complex JavaScript code in ServiceNow (Rejected due to maintenance complexity).
3. **ServiceNow Flow Designer (Selected):** A low-code triggerbased automation flow to create Catalog Tasks automatically upon request approval.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

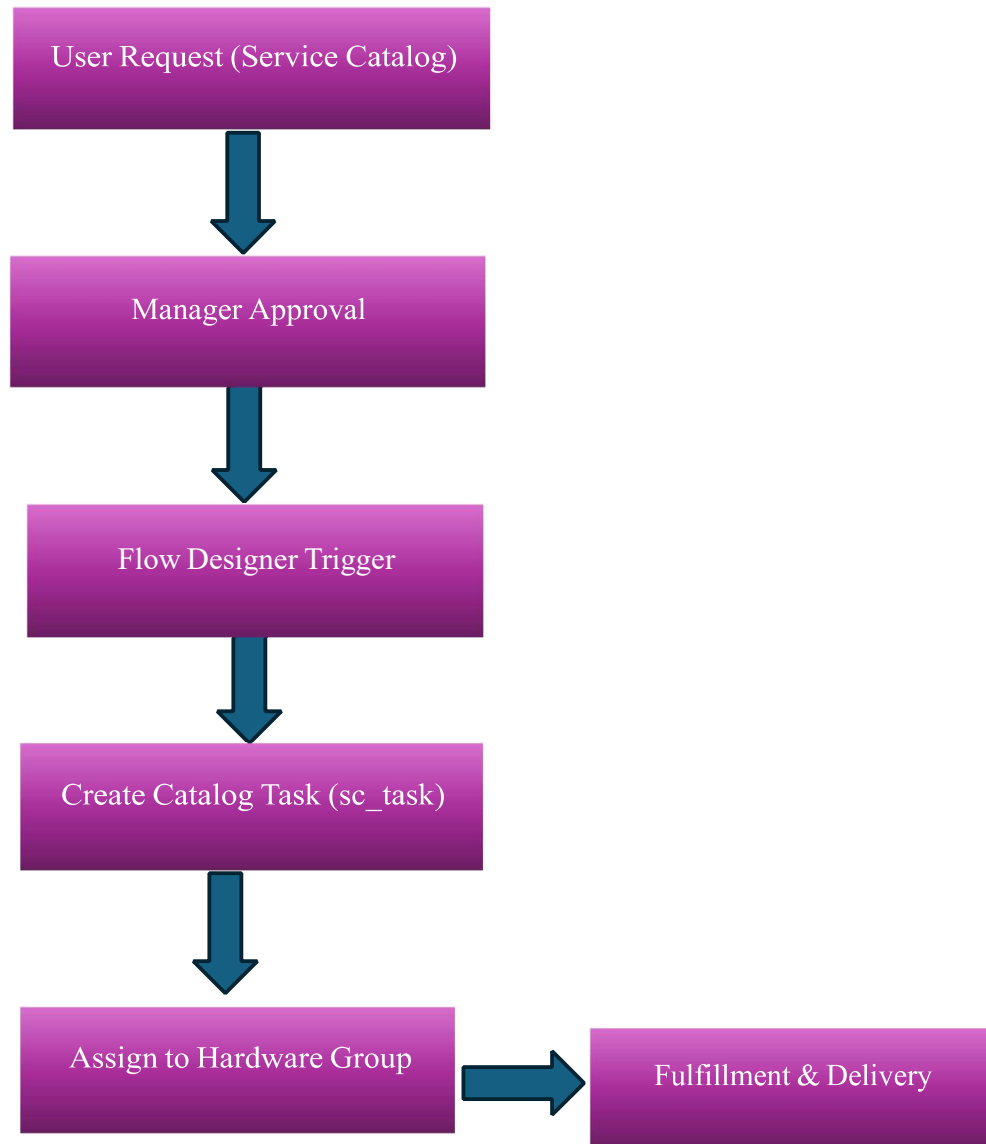
- **Entice:** User identifies the need for a laptop and searches the Service Catalog.
- **Enter:** User submits the standard laptop request via ServiceNow.
- **Engage:** Flow Designer triggers upon manager approval and generates a Catalog Task.

- **Exit:** IT Hardware team receives the auto-assigned task, configures the device, and delivers it.
- **Extend:** User receives the asset, closes the ticket, and submits feedback.

3.2 Solution Requirement

- Automated trigger on Service Catalog request approval.
- Automatic creation of a Catalog Task (sc_task).
- Pre-populated Short Description: "Laptop need to Configured".
- Direct assignment to the Hardware Assignment Group.

3.3 Data Flow Diagram



3.4 Technology Stack

- Platform: ServiceNow (Utah/Vancouver/Washington Release)
- Automation Module: ServiceNow Flow Designer
- Catalog Component: Service Catalog, Requested Item (sc_req_item), Catalog Task (sc_task)

4. PROJECT DESIGN

4.1 Problem Solution Fit

By replacing manual ticket processing with Flow Designer triggers, request fulfillment time is drastically reduced, ensuring that tasks are instantly made available to the Hardware team upon approval.

4.2 Proposed Solution

Implement an automated flow titled "Standard Laptop Task" using Flow Designer, attached to the Standard Laptop Catalog Item under Maintain Items.

4.3 Solution Architecture

- Trigger: Service Catalog
- Action: Create Catalog Task
 - Table: Catalog Task (sc_task)
 - Short Description: "Laptop need to Configured"
 - Assignment Group: Hardware
 - State/Approval: Approved

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- **Milestone 1:** Flow Creation in Flow Designer (Define Trigger, Create Action, Save & Activate).
- **Milestone 2:** Flow Assignment in Maintain Items (Attach flow to "Standard Laptop" item).
- **Milestone 3:** Service Catalog Testing (Order placement, approval, and task verification).

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

- Test Case 1: Submit Standard Laptop order → Result: Request created (REQ/RITM).
- Test Case 2: Manager approves request → Result: Flow executes in real-time.
- Test Case 3: Catalog Task creation check → Result: Task created with Short Description "Laptop need to Configured" and assigned to "Hardware" group successfully.

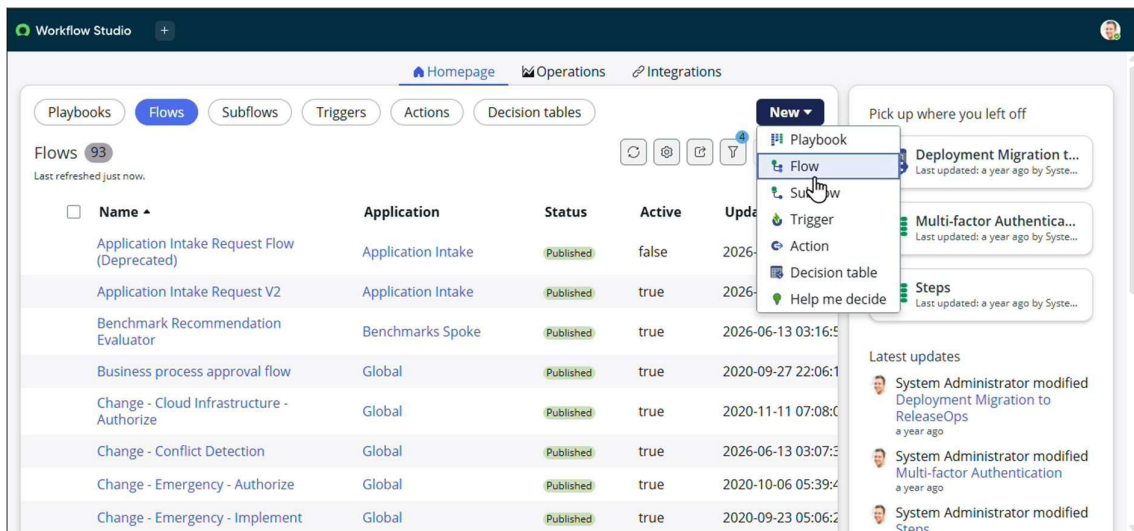
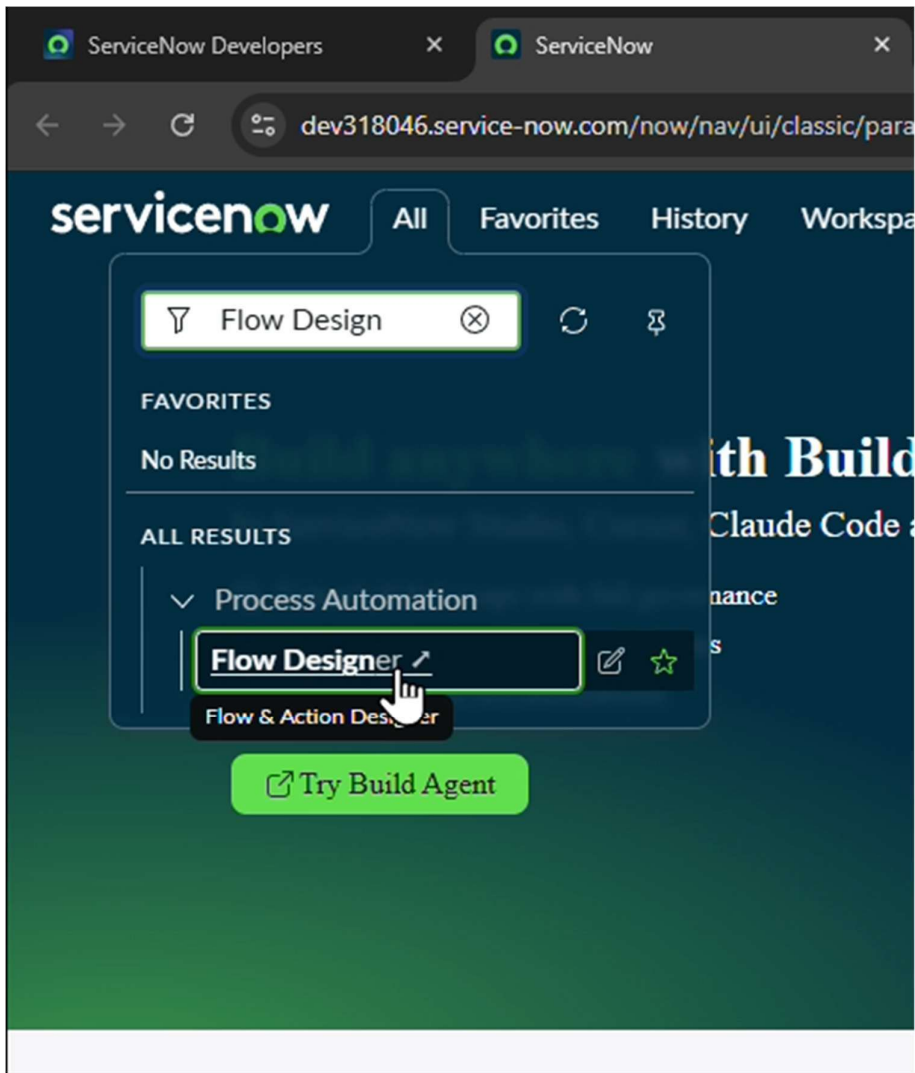
7. RESULTS

7.1 Output Screenshots & Execution Steps

MILESTONE 1: Flow Creation & Configuration Activity

1: Create Flow to Create Catalog Task

1. Open ServiceNow instance.
2. Click on All → Search for Flow Designer.
3. Click on Flow Designer under Process Automation.
4. Click on New → Select Flow.
5. Set Flow Name as "Standard laptop task".
6. Set Application as Global.
7. Select Run As → "System User".
8. Click Build Flow.



Workflow Studio New Flow Flow: None

Let's get the details for your flow

Name to uniquely identify your flow.

Flow name *

Application *

Description

[Show additional properties](#)

Activity 2: Configure Trigger & Action

9. Click on Add a Trigger → Search for "Service Catalog" → Click Done.
10. Under Actions, click Add an Action.
11. Search and select "Create Catalog Task".

Standard laptop task Inactive

View:

TRIGGER

Abort trigger creation

Trigger:

ACTIONS Select multiple

ERROR HANDLER ☐

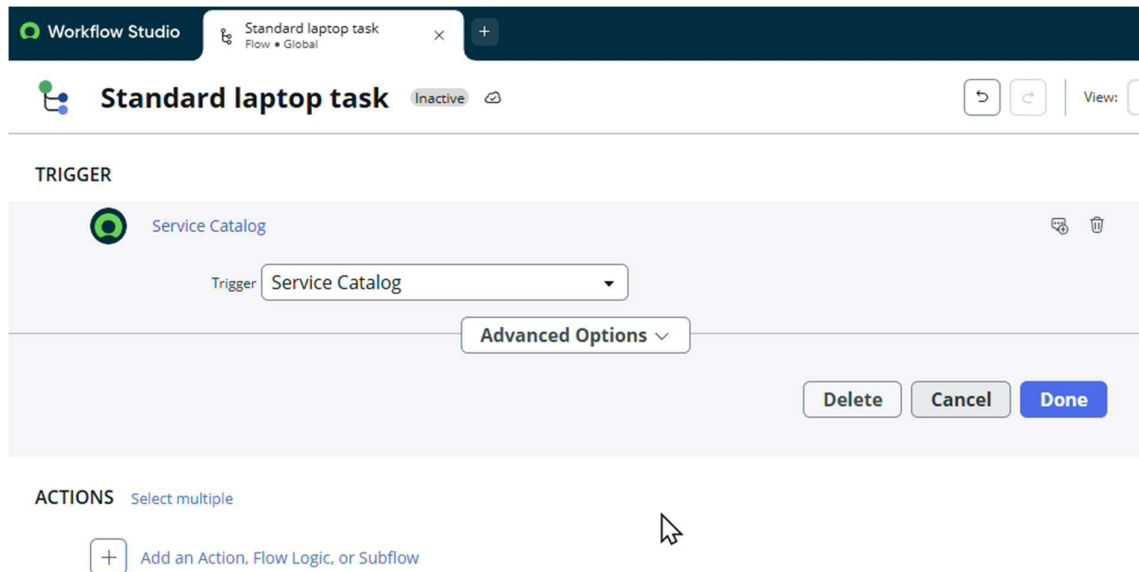
If an error occurs in your flow

Data

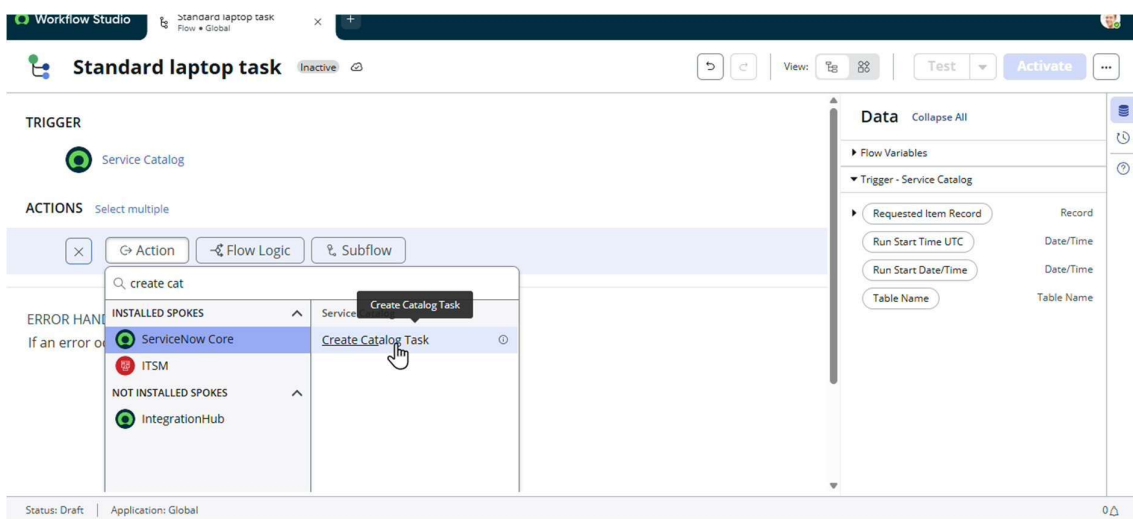
Flow Variables

Trigger

Status: Draft | Application: Global



12. Drag and drop Requested Item Record into the Request Item field.
13. Set Short Description as "Laptop need to Configured".
14. Set Description as "Laptop need to Configured".
15. Set Assignment Group as "Hardware".
16. Set Approval as "Approved".
17. Click Done → Click Save → Click Activate.



Standard laptop task Inactive

Table Name: **Catalog Task [sc_task]**

* Requested Item (Requested Item): **Trigger -> Requested Item ...**

Short Description: **Laptop need to Configured**

Fields:

- Description**: **Laptop need to Configured**
- Assignment group**: **Hardware**
- Approval**: **Approved**

+ Add field value

Wait: ☒

Template Catalog Item (Catalog Item): **Select Template Catalog Item**

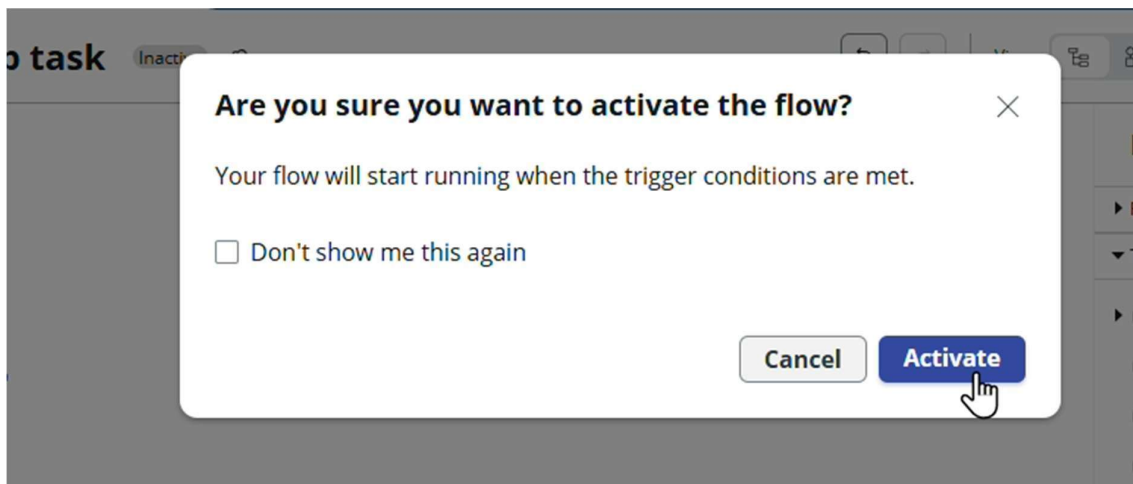
Catalog Variables:

- Available: **Select Template Catalog Item field**
- Selected: **Create Catalog Item(s) that**

Status: Modified | Application: Global

Data Collapse All

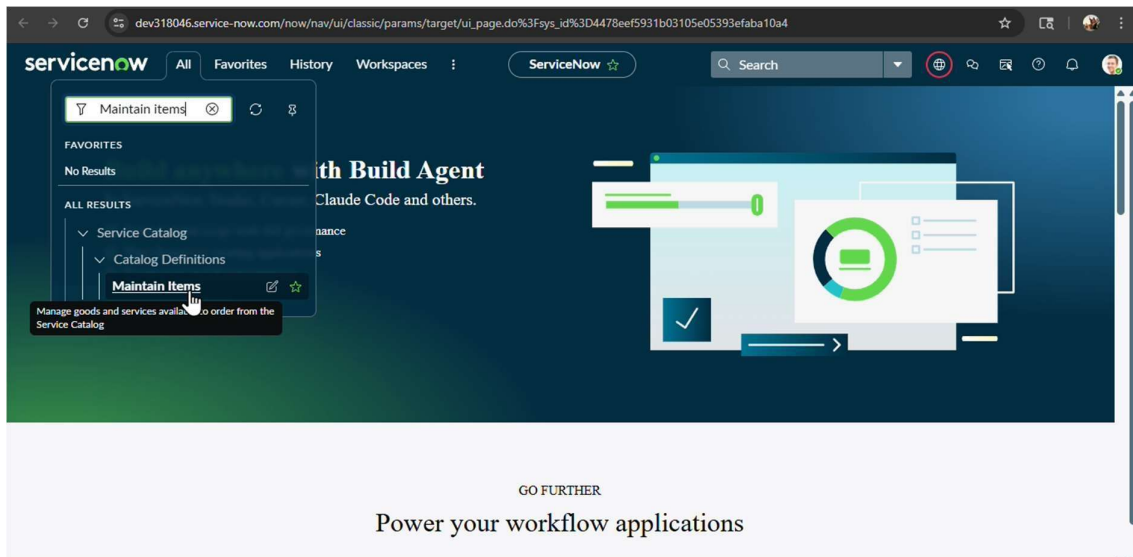
- Flow Variables
- Trigger - Service Catalog
 - Requested Item Record
 - Run Start Time UTC: Date/Time
 - Run Start Date/Time: Date/Time
 - Table Name: Table Name
- 1 - Create Catalog Task



MILESTONE 2: Flow Assignment to Catalog Item

Activity 1: Assign Flow to Standard Laptop Service Catalog

1. Click on All → Search for Maintain Items.
2. Select Maintain Items → Search for "Standard Laptop" record.
3. Under the record, select the Process Engine tab.
4. Attach the created flow "Standard Laptop Task" under the Flow field.
5. Click Save / Update to save the record.



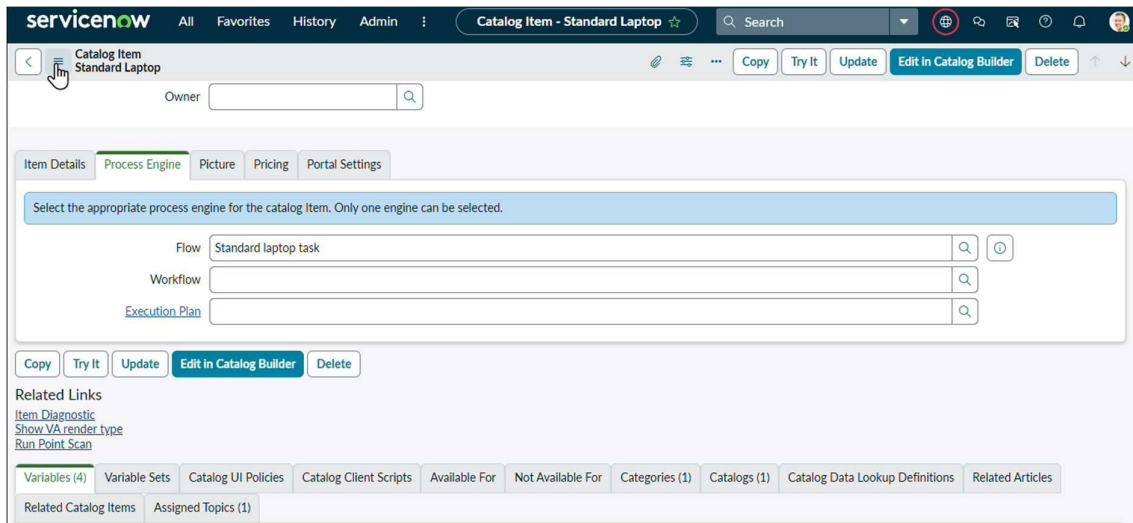
2. Select Maintain Items → Search for "Standard Laptop" record.

Catalog Items								
Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated
Standard Laptop	Lenovo - Carbon x1	true		Service Catalog	Hardware	\$1,100.00	Item	2022-11-20 20:46:33
StarTech Mini Display Port to DVI Adapter	PC / Mac Compatible	false		Service Catalog	Peripherals	\$17.23	Item	2022-11-20 20:46:33
StarTech Mini Display Port to VGA Adapter	PC / Mac Compatible	true		Service Catalog	Peripherals	\$24.50	Item	2022-11-20 20:46:33
StarTech USB Mini Hub	4 Port, USB 2.0	true		Service Catalog	Peripherals	\$11.55	Item	2022-11-20 20:46:33
StarTech USB to DVI Adapter	Multi Monitor Adapter	true		Service Catalog	Peripherals	\$49.99	Item	2022-11-20 20:46:33
STM MacBook Air Sleeve	STM Mac Sleeve - S13"	true		Service Catalog	Peripherals	\$24.26	Item	2022-11-20 20:46:33
STM MacBook Pro Sleeve	STM Mac Sleeve - S15"	true		Service Catalog	Peripherals	\$25.78	Item	2022-11-20 20:46:33
Table Index	Do you need to index a database table?	true		Service Catalog	Services	\$0.00	Item	2022-11-20 20:46:33

3. Under the record, select the Process Engine tab.

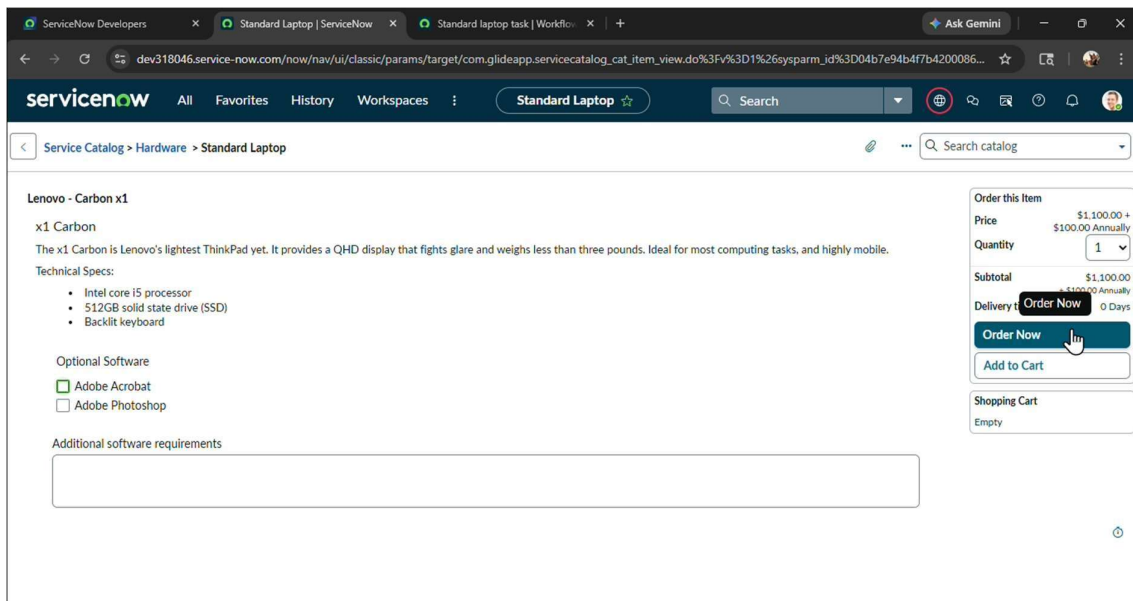
4. Attach the created flow "Standard Laptop Task" under the Flow field.

5. Click Save / Update to save the record.



Activity 1: Request Order & Verify Generated SCTASK

- Click on All → Search for Service Catalog → Select Hardware.
- Select Standard Laptop → Click Order Now to place the request.
- Click on the generated Request Number (REQ) → Open the Requested Item (RITM).



- Scroll down to the Approvers tab → Right-click and set approval state to Approved.
- Scroll down to the Catalog Tasks (SCTASK) section to verify automatic task creation.
- Open the SCTASK record and confirm:

o Short Description: "Laptop need to Configured" o

Assignment Group: "Hardware"

- It will open the Requested number record.
- Scroll down Click on the Approvers Section.

The screenshot displays the ServiceNow interface for Request REQ0010001. The top navigation bar includes 'Request - REQ0010001' and a search bar. Below the navigation bar, there are tabs for 'Request REQ0010001', 'Activities: 1', and 'Field changes • 2026-09-22 06:51:27'. The main content area shows the 'Approvers (1)' tab selected. Below this, there are buttons for 'Update', 'Add New Item', 'Cancel Request', 'Copy', and 'Delete'. A table titled 'Approval for = REQ0010001' is visible, with columns for 'State', 'Approver', 'Comments', and 'Created'. The table contains one row with the state 'Requested', approver 'Eric Schroeder', and creation date '2026-09-22 06:51:33'. A mouse cursor is pointing at the 'Requested' state in the table.

State	Approver	Comments	Created
Requested	Eric Schroeder		2026-09-22 06:51:33

- Right Click on the Record to see config in that Click on the Approve.

Request - REQ0010001

Activities: 1

System Administrator

Impact: 3 - Low
Opened by: System Administrator
Priority: 4 - Low
State: Open

Field changes • 2026-09-22 06:51:27

Update Add New Item Cancel Request Copy Delete

Requested Items (1)

State

Approval for = REQ

Approve

Reject

Prices (1)

Approver	Comments	Created
Eric Schroeder		2026-09-22 06:51:33

1 to 1 of 1

- Then click on the Requested item Section
- Open that Requested Item record

Requested Item - RITM0010001

Impact: 3 - Low
Opened by: System Administrator
Priority: 4 - Low
State: Open

Update Delete

Related Links

Flow Context

Catalog Tasks (1) Approvers Group approvals

Number Search

Request item = RITM0010001

Number	Assignment group	Assigned to	Short description	Actual start	Actual end	Order
SCTASK010001	Hardware	(empty)	Laptop need to Configured	2026-09-22 06:51:53	(empty)	

1 to 1 of 1

https://dev318046.service-now.com/sc_task.do?sys_id=77d06ea893a787505e0539...

- Scroll down to the “ Catalog tasks ” Section
- Open the record to see the updated status of the record with Short description and Assigned group.

The screenshot displays a ServiceNow Catalog Task form for the task ID SCTASK0010001. The form is titled "Catalog Task - SCTASK0010001" and includes a search bar and navigation links (Discuss, Follow, Update, Close Task, Delete). The form fields are organized into two columns. The left column contains: Number (SCTASK0010001), Assigned to (empty), Configuration item (empty), Active (checked), Short description (Laptop need to Configured), Description (Laptop need to Configured), and Work notes (Work notes). The right column contains: Approval (Approved), Priority (4 - Low), State (Open), Request item (RITM0010001), and Requested for (System Administrator). The bottom section shows Activities (1) and Field changes (2026-09-22 06:51:53).

8. ADVANTAGES & DISADVANTAGES

- **Advantages:**

- Eliminates manual delay in hardware task creation.
- Reduces manual errors and missed configurations.
- Improves transparency and tracking efficiency.

- **Disadvantages:**

- Requires correct initial catalog configuration.
- Workflow depends on predefined trigger logic.

9. CONCLUSION

By automating the standard laptop procurement process with ServiceNow's Flow Designer, the project effectively addresses inefficiencies and reduces manual overhead. This streamlined workflow ensures timely configuration of laptops, minimizes user wait times, and enhances overall productivity within the IT department.

10. FUTURE SCOPE

- Extending automated flow tasks for software allocation (e.g., MS Office, Adobe Suite).
- Integrating automated SMS/Teams notifications to endusers upon asset dispatch.

11. APPENDIX

- Source Code (if any): ServiceNow No-Code / Low-Code Flow (Flow Designer configuration)
- GitHub & Project Demo Link:
<https://github.com/addgjfj/Streamlining-IT-Procurement-Automating-Standard-Laptop-Orders-with-Flow-Designer.git>
- Project Demo Link-
<https://drive.google.com/file/d/1U3h3FvoBT2aydiZpTrTDwwtLq9eRNp4/view?usp=drivesdk>